

Assignment:

Exploration of Social Problems & Technology Opportunities

Question

What could become possible that is difficult or impossible with today's approaches?

Think beyond simply digitizing an existing process. Consider whether advances in AI, computer vision, sensing, speech, multimodal systems, optimization, robotics or other emerging technologies could fundamentally change **what can be done, who can do it, or the scale at which it can be done.**

Select 2 to 3 problem spaces that interest you from the list provided and explore the following:

1. Understand the problem

- a. Who experiences this problem?
- b. What does the problem look like on the ground?
- c. Why does the problem persist?
- d. Who is already trying to address it, and how?

2. Identify the technology opportunity

- a. What is currently difficult because of limitations of cost, scale, expertise, access or information?
- b. What has become possible because of recent advances in technology?
- c. Could AI, computer vision, sensing, speech, optimization, robotics or another emerging technology create meaningful leverage?
- d. What information, data or interaction would such a technology need?

3. Imagine—not design—the solution

- a. What could become possible if the technology worked well?
- b. Who would use it, and in what context?
- c. What would the technology do that existing approaches cannot do easily?
- d. What parts of the problem should remain human-led rather than automated?

Note: You are not expected to arrive at a final project idea. Instead, explore both the social problem and the potential technology opportunity.

Expected Output

Produce a short report with about one page per problem space explored, focusing on “problem understanding + potential technology + possible implications”. Avoid jumping to an App, feature list or future solution.

Problem Space

Potential Technology Direction

1	Detecting health conditions before they become serious Can cameras, mobile sensors or other low-cost devices enable screening or early detection outside hospitals and diagnostic centres?	Computer Vision + Sensing + Edge AI
2	Understanding what is happening in a person's health from multiple sources of information Can images, symptoms, questionnaires, longitudinal records, voice or other signals be combined to provide a more complete picture of an individual's health and risk?	Multimodal AI
3	Ensuring the right healthcare resource reaches the right person at the right time Can demand prediction, resource allocation and optimisation improve availability of blood, healthcare workers, screening camps or other scarce resources?	Prediction + Optimisation
4	Making essential information accessible across languages, literacy levels and abilities Can multilingual conversational systems, speech interfaces and language technologies make health, education and government information genuinely accessible?	NLP + Speech + Generative AI
5	Identifying what an individual learner needs to learn next Can AI identify learning gaps, misconceptions and learning needs and dynamically adapt content, assessment and feedback?	AI Personalisation + Intelligent Tutoring
6	Helping underserved people navigate the path from skills to meaningful livelihood Can technology understand a person's capabilities and circumstances and match them to appropriate training, jobs, mentors and opportunities?	Recommendation + Matching + Knowledge Graphs
7	Making conversations about gender, inclusion and sensitive social issues more effective and scalable How might an AI-based moderator support discussions by encouraging participation, identifying problematic or exclusionary interactions, surfacing different perspectives, asking probing questions, and helping facilitators manage the conversation?	AI Moderation + Conversational AI
8	Understanding and influencing behaviours that affect social outcomes Can data and AI help understand why people adopt—or fail to adopt—desired behaviours and enable interventions tailored to individuals or communities?	Behavioural Modelling + Personalisation
9	Helping frontline workers make better decisions with limited time and information Can AI collect information from multiple sources, identify cases requiring attention, recommend actions and assist field workers without replacing human judgement?	AI Decision Support + Agentic Systems
10	Enabling people with disabilities to interact with systems designed for everyone else Can computer vision, speech, sign-language technologies, gesture understanding and adaptive interfaces fundamentally change how people interact with digital services?	Accessibility Technology + Multimodal AI